YIHAN LI

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EDUCATION

Imperial College London **London, UK**

MSci Medicinal Chemistry (Four-Year Programme) — First-Class GPA:3.8/4 *Current*

Organic & Organometallic Chemistry: Reactivity at Carbon Centres; Chemistry of Molecular Systems; Control and Selectivity in Molecular Synthesis; Advanced Synthesis; Reactive Intermediates; Pericyclic Reactions; Process Chemistry

Inorganic & Structural Chemistry: Chemistry of the Elements: Hydrogen to Uranium; Structure and Bonding: Atomic Structure to Molecular Orbitals; Electronic States and Bonding

Physical & Computational Chemistry: Thermodynamics and Kinetics; Statistical Mechanics; Molecular Reaction Dynamics

Materials Chemistry: Solids, Liquids and Interfaces; Macromolecules and Materials; Soft Condensed Matter; Lyotropics; Membrane Biophysics; Nanomedicine

Medicinal Chemistry & Chemical Biology: Medicinal Chemistry; Strategies in Drug Discovery; Strategies in Cancer Chemotherapy; Introduction to Chemical Biology; From molecules to medicines

Analytical Chemistry & Laboratory Training: Introduction to Spectroscopy; Analysis of Molecules, Materials and Mixtures; Practical Chemistry I–III

Additional Academic Background: Economics, Microeconomics and Business; Korean Language (Levels 1–3)

RESEARCH EXPERIENCE

Research Assistant | Organometallic Chemistry & White Phosphorus Activation **2025**

Peking University

- Investigated the mechanism of white phosphorus activation by thioethers using complementary computational and experimental approaches; contributed to a manuscript currently in preparation.
- Investigated the reactivity of white phosphorus with Ti, Zr, Hf and Th cyclopentadienyl complexes, focusing on the structural, aromatic and electronic properties of diphosphabenzene species.
- Explored reactions of allenyl dilithium reagents with white phosphorus and subsequent product functionalisation.
- Performed computational analysis and compound characterisation using Gaussian/GaussView, ORCA, NMR, GC–MS, HRMS, XRD and elemental analysis.

Research Assistant | Solid-State Nanopore Protein Sensing **2026**

Zhejiang University

- Contributing to a research project on detecting proteins in tear-fluid samples using solid-state nanopore sensing.
- Conducting experimental work involving sample preparation, nanopore measurements and optimisation of experimental protocols.
- Developing practical experience in nanopore-based biosensing, electrical measurements and experimental troubleshooting.

Research Assistant | Supercapacitors Project **2024**

Xi'an Jiaotong University

- Conducted literature reviews on supercapacitor materials and energy-storage mechanisms, and assisted with electrode-material synthesis and characterisation using XRD and SEM.
- Supported preliminary analysis of electrochemical performance data and material-property comparisons.

MSci Research Project | Oxford Nanopore-Based Liquid Biopsy for Early Cancer Detection **2026-2027**

Imperial College London

- Investigating Oxford Nanopore Technologies (ONT)-based liquid-biopsy strategies for sensitive and multiplexed detection of low-abundance biomarkers for early-stage cancer detection.
- Developing and evaluating nucleic-acid probe systems that convert molecular recognition events into sequence-readable DNA barcodes for nanopore-based detection.
- Conducting experimental work in probe preparation, nanopore sequencing and assay optimisation, with a focus on miRNA and protein biomarker detection.

Research Project | Self-Healing Materials **2021**

Yuanpei College, Peking University

- Reviewed self-healing polymer mechanisms and applications and explored experimental approaches for evaluating material properties.
- Summarised findings and presented the project to peers and mentors.

ACADEMIC PROJECTS

Computational Exploration of Diels–Alder Reaction Mechanisms **2025**

Imperial College London

- Used Gaussian to perform geometry optimisation, transition-state searches, frequency calculations, IRC calculations and potential-energy-surface scans.
- Compared PM6 and B3LYP/6-31G(d) methods for reaction barriers, selectivity and mechanistic interpretation, including frontier-molecular-orbital and kinetic-versus-thermodynamic analysis.
- Troubleshot transition-state and IRC convergence using constrained optimisation, PES scans and adjusted computational settings.

Medicinal Chemistry & Drug Discovery Projects **2025–2026**

Imperial College London

- Applied SAR and physicochemical-property analysis to design P7-based antileishmanial acyl urea analogues; synthesised and characterised target compounds using multistep organic synthesis, chromatography, NMR, IR, MS/HRMS and HPLC.
- Performed structure-based design of SARS-CoV-2 Mpro inhibitors using MOE and PyMOL, including ligand redocking, binding-mode analysis, fragment growing and analogue optimisation, with proposed synthetic and FRET-based validation strategies.
- Analysed the development of SGLT2 inhibitors from phlorizin to dapagliflozin to understand how lead optimisation balances potency, selectivity, metabolic stability, oral bioavailability and pharmacokinetics.
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ENTREPRENEURSHIP & TEACHING

Co-Founder & Tutor | Education Consulting and Tutoring **2023–Present**

Self-Employed

- Co-founded and operate an education consultancy providing A-Level/Foundation tutoring, interview preparation and university application support.
- Manage client acquisition and social-media marketing while designing personalised Chemistry, Physics and Mathematics teaching materials.
- Secured multiple paid clients and delivered tailored academic and admissions support.

TECHNICAL SKILLS

Computational Chemistry: Gaussian/GaussView, ORCA (DFT calculations), MOE, PyMOL, LAMMPS (molecular dynamics)

Programming & Data: Python (data analysis and scripting), MATLAB; experimental data analysis and interpretation

Laboratory & Characterisation: Organic synthesis; nanopore measurements; NMR, GC–MS, HRMS, XRD, SEM, IR, HPLC; chromatography

Professional: Research troubleshooting, scientific communication, client management, teaching, business development and teamwork